

Splines: flexible models for nonlinear effects

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Overview

Categorization

Interpolating splines

Smoothing splines

Splines in R

Introduction

- Splines are a flexible class of models that can be helpful for representing dose-response relationships in epidemiology.
- In this course we will be using spline models extensively.
- However, spline models are widely misunderstood.
- The purpose of this lecture is to give a conceptual background on where spline models come from.

Outline

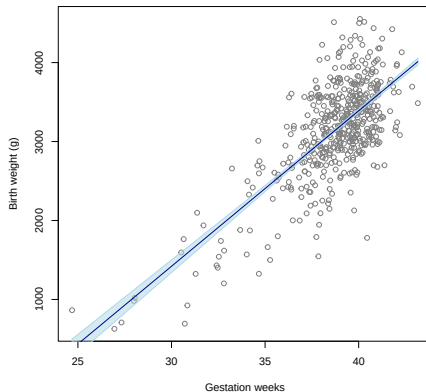
Categorization

Interpolating splines

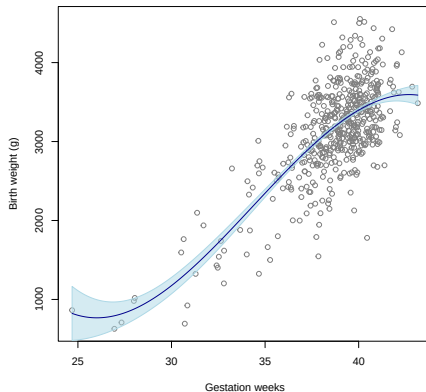
Smoothing splines

Splines in R

Predicting birth weight from gestational age



A cubic model for the births data

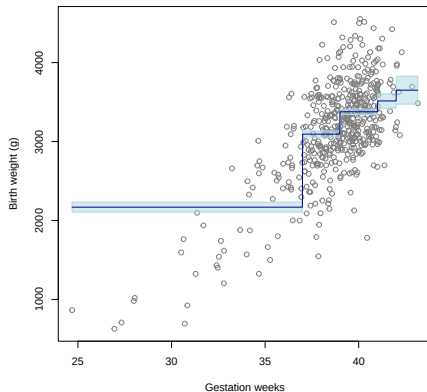


Categories

Medical doctors like to think in terms of categories

- **preterm** < 37 weeks
- **early term** 37-39 weeks
- **full term** 39-41 weeks
- **late term** 41-42 weeks
- **post term** ≥ 42 weeks

Fitting a categorical model

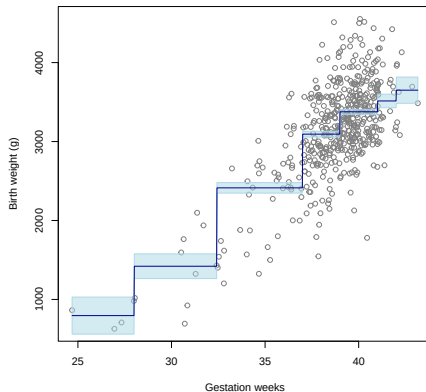


More categories

The poor fit for the category “preterm” can be improved by adding more categories:

- **extremely preterm** < 28 weeks
- **very preterm** 28-32 weeks
- **moderate to late preterm** 32-37

Plotting a more detailed categorical model



Statisticians against categorization

- Greenland S (1995) Avoiding power loss associated with categorization and ordinal scores in dose-response and trend analysis, *Epidemiology*, **6**, 450–454.
- Senn S (2005) Dichotomania: an obsessive compulsive disorder that is badly affecting the quality of analysis of pharmaceutical trials.
- Bennette C, and Vickers A, (2012), Against quantiles: categorization of continuous variables in epidemiologic research, and its discontents. *BMC Medical Research Methodology* 12:21

Epidemiologists against categorization

Rose, G. (1992) The Strategy of Preventive Medicine

- Many diseases are not discrete. Instead there is an underlying continuum of increasing severity.
- In medicine, we tend to conflate a clinical action (treat vs. do not treat) with the presence/absence of disease.
- Disease prevention efforts are best targeted at shifting the distribution of risk for the whole population instead of trying to identify and target a “high risk” group.

Example: hypertension

In 2017, the American College of Cardiology & American Heart Association changed the definition of hypertension

old definition $\geq 140/90$

new definition $\geq 130/80$

This increased the estimated prevalence of hypertension in the USA from 32% to 46% adding 31 million new cases ¹

¹Muntner et al 2018: <https://pubmed.ncbi.nlm.nih.gov/29133599/>

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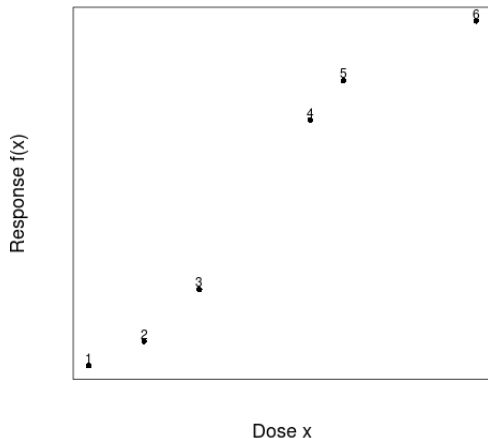
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Join the dots



- Suppose that we have a set of (x, y) points that we think come from an underlying smooth relationship between x and y .
- We want to join the dots in a way that is as smooth as possible.
- This turns out to be a mathematically well defined problem with a unique solution.

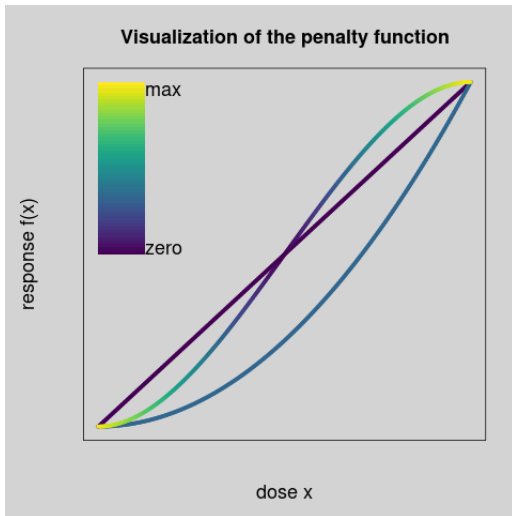
A roughness penalty

- Suppose $y = f(x)$ for some function $f(\cdot)$.
- The *roughness* of the curve in the interval $[a, b]$ is measured by the integral

$$\int_a^b \left(\frac{\partial^2 f}{\partial x^2} \right)^2 dx$$

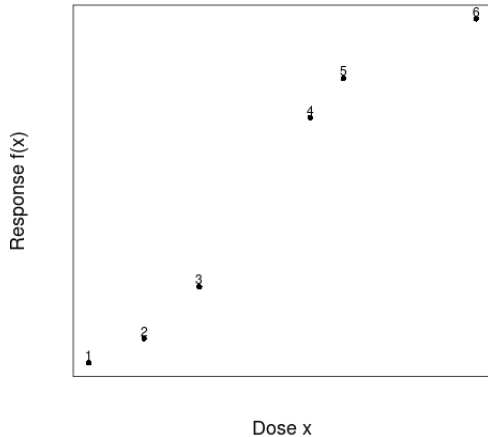
- We want the roughness of f to be as small as possible.

What does the roughness penalty mean?



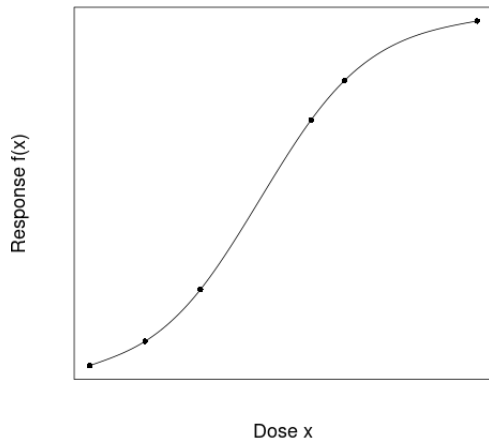
- The contribution to the penalty at each point depends on the curvature (represented by a colour gradient)
- A straight line has no curvature, hence zero penalty.
- Sharp changes in the slope are heavily penalized.

An interpolating cubic spline



- The smoothest curve that goes through the observed points is a cubic spline.

An interpolating cubic spline



- The smoothest curve that goes through the observed points is a cubic spline.

What is a cubic spline?

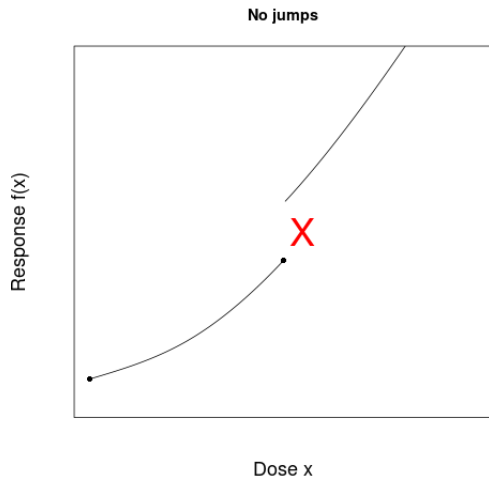
Splines are piecewise cubic curves

- There are fixed points on the x-axis called *knots*
- The knots divide the curve into sections
- Each section is a cubic function

$$f(x) = a + bx + cx^2 + dx^3$$

- The parameters a, b, c, d are different for different sections

Boundary conditions

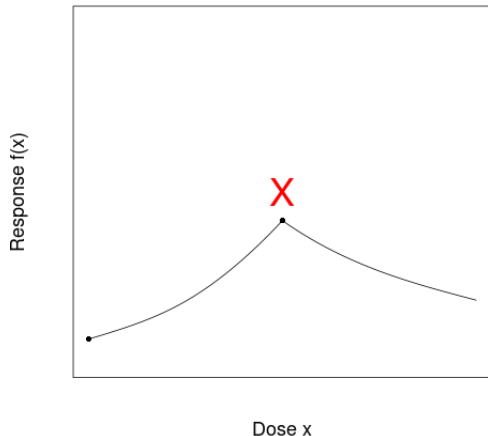


Sections need to join up smoothly.

- Both sides must go through the knot.
- The slope cannot change at a knot
- The curvature cannot change at a knot

Boundary conditions

No corners

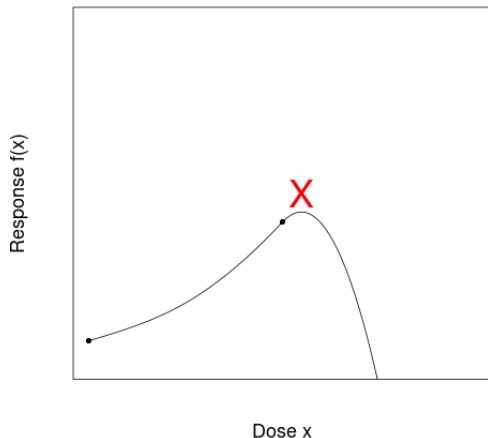


Sections need to join up smoothly.

- Both sides must go through the knot.
- The slope cannot change at a knot
- The curvature cannot change at a knot

Boundary conditions

No sudden changes in curvature



Sections need to join up smoothly.

- Both sides must go through the knot.
- The slope cannot change at a knot
- The curvature cannot change at a knot

Outline

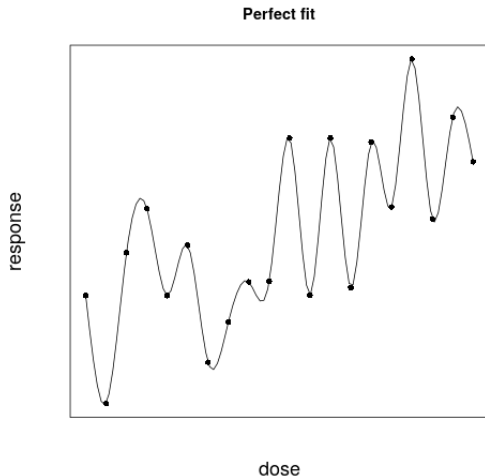
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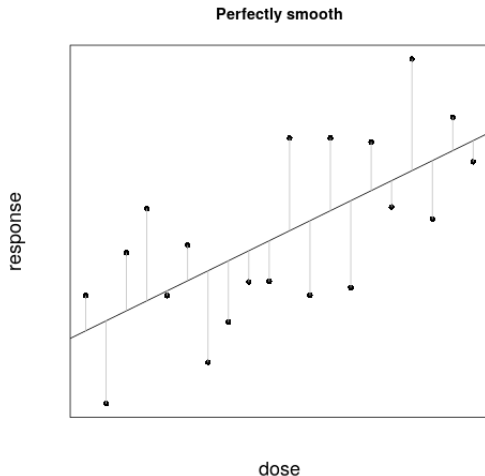
Dose response with error



In practice we never know the dose response curve exactly at any point but always measure with error.

An interpolating spline is not useful in this case. It fits the curve to the noise as well as the underlying dose-response.

Dose response with error



We know how to fit a maximally smooth (linear) curve to data with errors with linear regression. Linear regression assumes

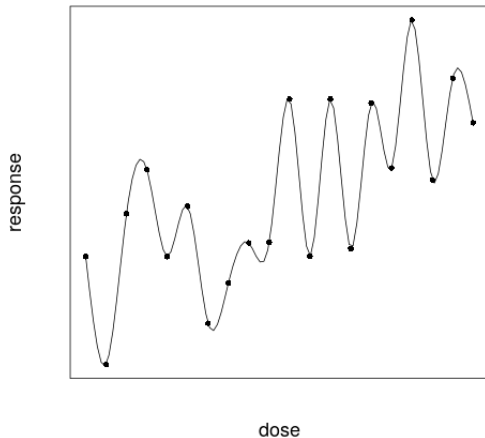
$$f(x) = \alpha + \beta x$$

Then we estimate α, β by minimizing the sum of squares

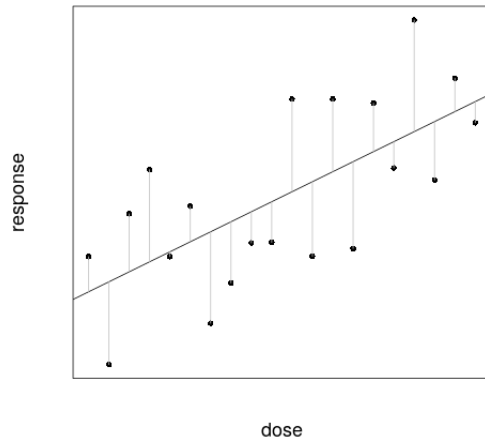
$$\sum_i [y_i - f(x_i)]^2$$

Maximum fit vs Maximum smoothness

Perfect fit



Perfectly smooth



Fitting a smoothing spline

Assuming $f(x)$ is a spline with fixed knots, minimize

$$\sum_i [y_i - f(x_i)]^2 + \lambda \int \left(\frac{\partial^2 f}{\partial x^2} \right)^2 dx$$

Or, more generally

Deviance $+ \lambda \times$ Roughness penalty

Size of tuning parameter λ determines compromise between model fit (small λ) and smoothness (large λ).

Smoothing and degrees of freedom

Software will choose the smoothing parameter λ for you automatically using cross-validation.

The smoothing parameter is adapted to the data.

Smoothness of the model can be measured with the *effective degrees of freedom* (EDF)

- Linear model: maximally smooth
 - EDF=2 (intercept + slope parameter)
- Interpolating mode: best fit
 - EDF=n (one parameter for every observation)

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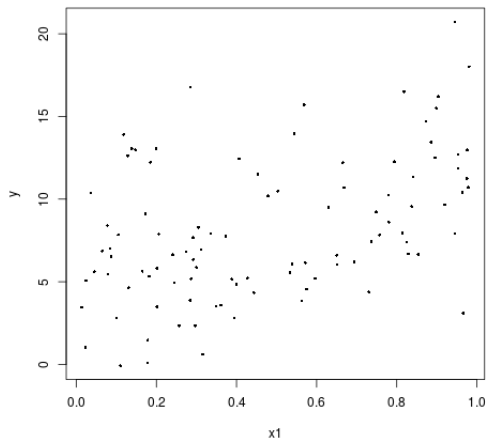
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Spline models in R

- Do not use the `splines` package.
- Use the `gam` function from the `mgcv` package to fit your spline models.
- The `gam` function chooses number and placement of knots for you and estimates the size of the tuning parameter λ automatically.
- You can use the `gam.check` function to see if you have enough knots. Also re-fit the model explicitly setting a larger number of knots (e.g. `double`) to see if the fit changes.

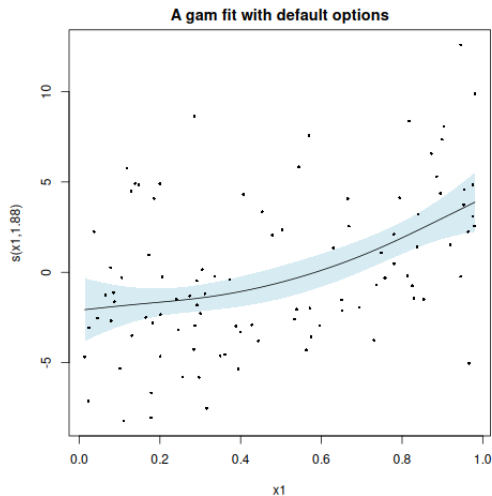
Penalized spline

Some simulated data



- A gam fit to some simulated data
- Model has 9 degrees of freedom
- Smoothing reduces this to 1.88 effective degrees of freedom

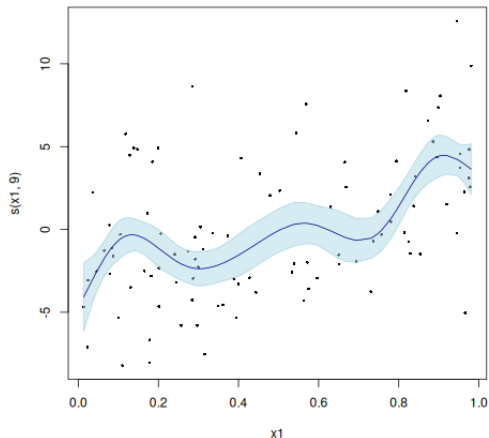
Penalized spline



- A gam fit to some simulated data
- Model has 9 degrees of freedom
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Unpenalized spline

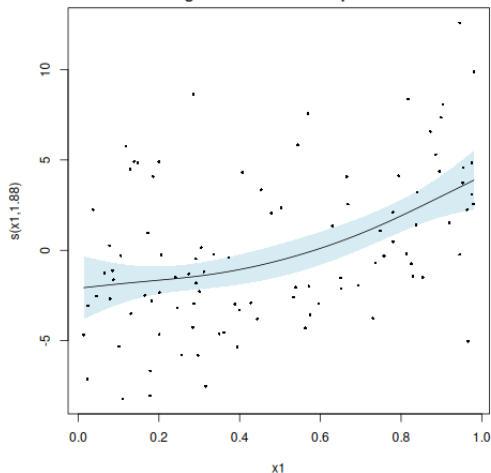
An unpenalized spline fit



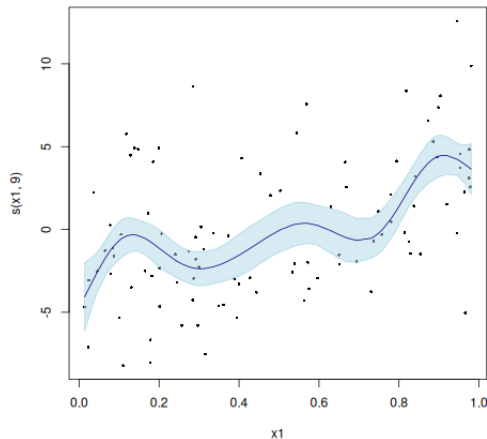
- An unpenalized spline using the same spline basis as the gam fit.
- Model has 9 degrees of freedom

Penalized vs unpenalized splines

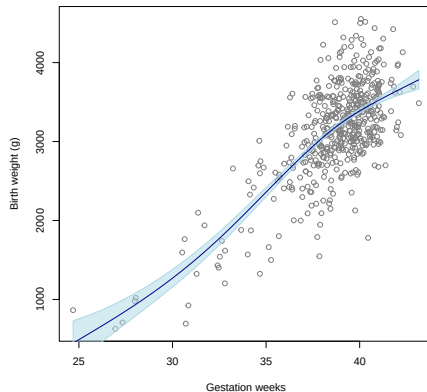
A gam fit with default options



An unpenalized spline fit



A penalized spline model for the births data



Conclusions

- Epidemiologists like to turn continuous variables into categories.
- Statisticians do not like categorization because it loses information.
- Splines are a flexible class of models that avoid categorization but also avoid making strong assumptions about the shape of a dose-response relationship.
- Penalized regression splines are based on compromise between goodness-of-fit and smoothness.
- Most of the decisions in fitting a penalized regression spline can be made for you
 - Degree of smoothing
 - Number of knots
 - Placement of knots